Sharpness of the Avoid-or-Compress Bound for Reachable Images

Qichao Wang
Hebei University of Technology

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Abstract

Let $\mathrm{ac}(S, A)$ be the minimum word length omitting a state of $\varnothing \neq A \subsetneq S$ or compressing S. We prove sharpness of Szykuła's bound $n - |A|$ on the first two cardinality faces (proper S on the second) and at every $n \geq 2|S| - |A| - 1$ for larger gaps. All witnesses are binary, strongly connected and synchronizing, with S a reachable image. Put $m = n - |S|$ and $q = n - |A|$. Both binary and unrestricted maxima equal one for $m = 0$, $\min\{q, 3m\}$ for $1 \leq m \leq 3$, and q for $m \geq 4$, $q \leq 3m - 1$. The codimension-three upper uses a computer-assisted structural proof and refutes $\min\{q, \binom{m+2}{2}\}$ at infinitely many parameters. Independently checked witnesses determine every parameter through $n = 12$. The remaining region is unresolved; this is a partial answer to Szykuła's 2026 Open Problem 4.

1 Statement and preliminaries

A complete deterministic automaton is a triple (Q, Σ, δ) with finite state set Q and total transition function $\delta : Q \times \Sigma \to Q$. We use right actions: $qw = \delta(q, w)$ and $Sw = \{qw : q \in S\}$. Length counts letters of Σ. The automaton is strongly connected if every state reaches every other state, and synchronizing if $|Qw| = 1$ for some word w. A subset S is a reachable image if $S = Qv$ for a word v, possibly empty. Put

$$\mathrm{ac}(S, A) = \min\{|w| : A \nsubseteq Sw \text{ or } |Sw| < |S|\}, \qquad \varnothing \neq A \subsetneq S \subseteq Q.$$

Here avoidance means omitting at least one state of A, not all of A. Call a word safe if $A \subseteq Sw$ and $|Sw| = |S|$.

The upper bound $n - |A|$ is due to Szykuła [1, Lemma 2]; it is restated as Lemma 3 in his 2026 survey [2]. Open Problem 4 asks for tight bounds depending on both cardinalities. Figure 2 of that survey already gives the case $(|S|, |A|) = (2, 1)$. Ferens, Szykuła and Vorel [3, p. 46:6] also discuss sharpness for a related fixed-state avoiding-or-compressing alternative. We make no priority claim for these earlier cases. Here we give explicit cardinality-dependent witnesses with all the structural requirements imposed simultaneously.

Let $\mathcal{F}(n, s, r)$ denote the maximum of $\mathrm{ac}(S, A)$ over strongly connected synchronizing n-state automata with reachable image S, $|S| = s$ and $|A| = r$. Write $\mathcal{F}_{\mathrm{bin}}$ for its restriction to two-letter alphabets. The avoid-or-compress lemma and the Frankl–Pin compression bound [1, Theorem 1] give

$$\mathcal{F}_{\mathrm{bin}}(n, s, r) \leq \mathcal{F}(n, s, r) \leq U(n, s, r) := \min\{n - r, \tbinom{n-s+2}{2}\}.$$

Theorem 1. *The following bounds hold, and every lower bound has a binary witness.*

1. $\mathcal{F}(n, r+1, r) = n - r$ *for* $n \geq 2$ *and* $1 \leq r \leq n - 1$.
2. $\mathcal{F}(n, r+2, r) = n - r$ *for* $n \geq 4$ *and* $1 \leq r \leq n - 3$.
3. *For* $r \geq 1$, $s - r \geq 3$ *and* $n \geq 2s - r - 1$, *one has*

$$\mathcal{F}(n, s, r) = n - r.$$

For $1 \leq r < n$, *one has* $\mathcal{F}(n, n, r) = 1$.

Thus the first two cardinality faces are determined, including their full-image boundary. For every fixed (s, r) with $s - r \geq 3$, the bound is sharp at every $n \geq 2s - r - 1$, in particular at $n = r + k(s - r - 1) + 2$ for $k \geq 2$.

We recall the upper bound with a proof, to make the argument self-contained. For a word w let $D_w(q, t) = 1$ if $qw = t$, and 0 otherwise. Then $D_{uv} = D_u D_v$. The characteristic vector χ_S is a row vector; $\chi_S D_w$ counts trajectories from S, with multiplicity, and its support is Sw.

Lemma 2 (Szykuła [1, 2])**.** *If $A \not\subseteq Sz$ for some word z, then* $\mathrm{ac}(S, A) \le n - |A|$.

Proof. Write $r = |A|$, $s = |S|$, and let V_i be the span over $\mathbb{R}$ of $\chi_S D_w$ for $|w| \le i$. Suppose all words of length at most $n - r$ are safe. Safety gives $|Sw| = |S|$, so $S \to Sw$ is injective: no trajectories collide, and every image state has multiplicity 1. Their rows therefore have total s and value 1 at every coordinate of A. Hence V_{n-r} lies in

$$H = \{x : x_q = s^{-1} \textstyle\sum_{t \in Q} x_t \text{ for every } q \in A\}.$$

The r equations are independent since $r < s$, so $\dim H = n - r$. Now $\dim V_0 = 1$, and $V_i = V_{i+1}$ implies invariance under every letter and stabilization forever. Therefore some equality must occur before index $n - r$, or $\dim V_{n-r} \ge n - r + 1$. Stabilization puts $\chi_S D_z$ in H, contradicting its zero coordinate in A and total s. □

In a strongly connected synchronizing automaton with $n \ge 2$, a reset word followed by a path to a state outside A gives the required z. This proves every upper bound $n - r$ in Theorem 1. If $S = Q$, some letter must be nonpermutational and immediately compresses Q, while the empty word is safe. Thus $\mathrm{ac}(Q, A) = 1$.

We will also use the following elementary fact. If t cycles $x_0, x_1, \ldots, x_{s-1}$ and e sends x_0 to x_1 while fixing the other points, then

$$e(t^{s-1}e)^{s-2} \tag{1}$$

maps this set to x_1. Indeed, after e the image is $\{x_1, \ldots, x_{s-1}\}$; each $t^{s-1}e$ deletes the last point of the remaining interval. This also covers $s = 2$.

2 The first two faces

Proposition 3. *Part (1) of Theorem 1 holds.*

Proof. Put $L = n - r \ge 1$ and $Q = \{0, \ldots, n - 1\}$. Define

$$qa = \begin{cases} q + 1 & q < L, \\ q & q \ge L, \end{cases} \qquad S = \{0, L, L + 1, \ldots, n - 1\}, \qquad A = S \setminus \{0\}.$$

Let b cycle S in the displayed order and send every point outside S to L. Then $Qb = S$. The a-path from 0 to L and the b-cycle prove strong connectivity.

For $0 \le j < L$, set $S_j = A \cup \{j\}$. One has $S_j b = S_0$, and $S_j a = S_{j+1}$ for $j < L - 1$; $S_{L-1} a = A$. Induction shows that every word of length less than L has image S_j with $j < L$, so is safe. The word a^L compresses S, proving $\mathrm{ac}(S, A) = L$.

On S, $c = a^L$ is the unit $0 \mapsto L$. Formula (1) with $t = b$ shows that $c(b^r c)^{r-1}$ resets S. It resets Q as well, since $Qc = A = Sc$. This proves synchronization, also when $L = 1$ or $r = 1$. □

Proposition 4. *Part (2) of Theorem 1 holds.*

Proof. Put $M = n - r \ge 3$. Take disjoint sets $A = \{p_0, \ldots, p_{r-1}\}$ and $U = \{u_1, \ldots, u_M\}$, and let $S = A \cup \{u_1, u_2\}$. Letter a fixes A, while

$$p_i b = p_{i+1} \quad (0 \le i < r - 1), \qquad p_{r-1} b = u_1.$$

For odd M, let a cycle $u_1, \ldots, u_M$ and set

$$u_j b = \begin{cases} p_0 & j \text{ odd}, \\ u_2 & j \text{ even}. \end{cases}$$

For even M (necessarily $M \ge 4$), set

$$u_1 a = u_2, \quad u_j a = u_{j+1} \ (2 \le j < M), \quad u_M a = u_2, \qquad u_j b = \begin{cases} u_2 & j = 1 \text{ or } j \ge 3 \text{ odd}, \\ p_0 & j \text{ even}. \end{cases}$$

In both cases $Qb = S$. From p_0 the b-path reaches u_1, whose a-orbit covers U; some state of U returns to p_0 under b. Every state therefore reaches and is reached from p_0.

The two moving points of Sa^j remain distinct for all $j \ge 0$. For $0 \le j \le M - 2$, their b-images are p_0 and u_2, so $Sa^j b = S$. At $j = M - 1$, the pair is $\{u_M, u_1\}$ for odd M and $\{u_M, u_2\}$ for even M; both points map to p_0. Thus $h = a^{M-1} b$ compresses S. Every word shorter than M splits into a-runs followed by b, each run having length at most $M - 2$, and a trailing a-run. All these transitions are safe. Hence $\mathrm{ac}(S, A) = M$.

Set $s = r + 2$, and $t = ab$ for odd M, $t = b$ for even M. On S, t cycles $p_0, \dots, p_{r-1}, u_1, u_2$. If e is the unit $u_1 \mapsto u_2$ on S, then $h = et$ there. Consequently

$$h(t^{s-2}h)^{s-2} = e(t^{s-1}e)^{s-2}t$$

on S, and this is a reset by (1). Prepending b gives a reset of Q. The unit e is used only to verify the identity; the reset word uses the original letters. □

3 Every larger fixed gap

Proposition 5. *For $r \geq 1$, $g \geq 2$ and $M \geq 2g$, there is a binary strongly connected synchronizing automaton with*

$$n = r + M + 1, \quad |S| = s = r + g + 1, \quad |A| = r, \qquad \mathrm{ac}(S, A) = M + 1.$$

The subset S is an actual reachable image.

Proof. Take disjoint sets $P = \{p_0, \dots, p_r\}$ and $U = \{u_1, \dots, u_M\}$. Set $A = P \setminus \{p_r\}$ and $S = P \cup \{u_1, \dots, u_g\}$. First suppose $M \geq 2g + 1$ and put $h = 1 + ((M - 1) \bmod g)$. Then $1 \leq h \leq g$ and $M \geq 2g + h$. Let a fix P, send u_j to u_{j+1} for $j < M$, and send u_M to u_{h+1}. Define $p_0 b = u_h$, $p_i b = p_{i-1}$ for $1 \leq i \leq r$, and

$$u_j b = \begin{cases} p_r & j \equiv h - 1 \pmod g, \\ u_1 & j = g,\ h \geq 2, \\ u_{h+g} & j = M, \\ u_{j+1} & \text{otherwise.} \end{cases}$$

The cases are disjoint. On S, b cycles $p_0, u_h, \dots, u_g, u_1, \dots, u_{h-1}, p_r, \dots, p_1$. The segment $u_1, \dots, u_{h-1}$ is empty for $h = 1$. Ordinary outside states follow increasing indices to the next index congruent to $h - 1$, then enter p_r. The exceptional path is $u_M, u_{h+g}, \dots, u_{h+2g-1}, p_r$, of length $g + 1$. Its indices exist because $M \geq 2g + h$. Thus $Qb^{g+1} = S$. The a-path from u_1 covers U, and the b-cycle covers P, proving strong connectivity.

Safety. Use singleton P-cells and color cells $1, \dots, g$, assigning u_j color $1 + ((j - h - 1) \bmod g)$. Let C be the $n \times s$ state-to-cell incidence matrix. Let T_a fix P and cycle the colors, and let T_b cycle $p_0, g, 1, \dots, g - 1, p_r, \dots, p_1$. For a standard state column f_q and cell row e_c, one has

$$D_a C = C T_a, \qquad D_b C = C T_b + f_{u_M}(e_g - e_1). \tag{2}$$

The last term is an outer product of an $n \times 1$ column and a $1 \times s$ row. These cells are bookkeeping devices, not a quotient automaton. For $w = \ell_1 \cdots \ell_t$, iteration gives

$$\chi_S D_w C = \mathbf{1}_s + \sum_{\substack{1 \leq i \leq t \\ \ell_i = b}} \mu_i (e_g - e_1) T_{\ell_{i+1} \cdots \ell_t}, \qquad \mu_i = (\chi_S D_{\ell_1 \cdots \ell_{i-1}})_{u_M} \geq 0. \tag{3}$$

This counts trajectories even after collisions, with products in word order. A trajectory from S needs at least $M - g$ letters to reach u_M: initial indices are at most g, every transition within U increases an index by at most one, and entrances from P have index $h \leq g$. Hence a nonzero correction has $i \geq M - g + 1$. In the cell graph, color g needs at least $g - 1$ letters to reach color $g - 1$, and color 1 needs at least g letters to reach A. The latter route traverses $1, \dots, g - 1$ and then takes two letters b through p_r. A detour through P cannot shorten either route.

For $t \leq M - 1$, correction suffixes have length at most $g - 2$, so no positive correction reaches color $g - 1$, whose trajectory count is at most one. Suppose a word of length at most M has a first collision. For a colliding b, the distinct source states must lie in its only non-singleton fiber, all of color $g - 1$, contradicting this count bound. For a colliding a, the sources must be u_h, u_M, its only non-singleton fiber. They are not both initially present. A preceding b cannot give u_M, which has no b-preimage. A preceding a requires sources u_{h-1}, u_{M-1} one step earlier, both of color $g - 1$, again a contradiction; for $h = 1$, u_h has no a-preimage at all. Thus every word of length at most M preserves s states. For $t \leq M$, every correction suffix has length at most $g - 1$, so no negative correction reaches A. Its singleton cells retain multiplicity at least one. Every such word is safe.

The word $a^{M-g}b^{g+1}$ has length $M + 1$. Exactly its first b has a trajectory at u_M, with multiplicity one. Its negative correction is $e_1 T_b^g = e_{p_{r-1}}$, whereas its positive correction is $e_g T_b^g = e_{p_r}$. It therefore omits p_{r-1} and attains equality, also when $r = 1$.

For synchronization put $c = a^{M-h}b^s$. The a-run fixes the cyclic tail $u_{h+1}, \dots, u_M$. For $j < h$, it sends u_j to u_{M-h+j}, which has the same remaining b-distance $h - j$ to p_r as u_j. For $j = h$, it sends u_h to u_M, which reaches p_0 in $g + 1 + r = s$ letters b. Consequently c fixes $S \setminus \{u_h\}$ and sends u_h to its b-predecessor p_0. After c, each bc extends the missing interval by one point in forward b-order. Thus $b^{g+1}c(bc)^{s-2}$ resets Q.

The boundary $M = 2g$. Let a fix P and cycle U. Define $p_0 b = u_g$, $p_i b = p_{i-1}$ and

$$u_j b = \begin{cases} p_r & j \equiv g-1 \pmod g, \\ u_{j+1} & j < M,\ j \not\equiv g-1 \pmod g, \\ u_M & j = M. \end{cases}$$

Now b cycles $B = P \cup \{u_g, \dots, u_{2g-1}\}$ and fixes u_{2g}; $Qb^{g+1} = E = B \cup \{u_{2g}\}$. Put $d = a^{g-1}b^s$. It fixes P and on E has actions

$$u_g d = u_{2g-1}, \quad u_{g+1}d = u_{2g}, \quad u_{g+j}d = u_{g+j-1}\ (2 \le j \le g-1), \quad u_{2g}d = u_{2g-1}.$$

Therefore $Eda^g = S$, proving actual reachability. The two cycles prove strong connectivity. Assigning u_j color $1 + ((j-1) \bmod g)$ gives exactly (2)–(3) again. The same delay and collision arguments apply, with a now a permutation. The word $a^g b^{g+1}$ attains avoidance: only the first negative correction has a suffix of g letters b and reaches p_{r-1}; no positive correction reaches A. Finally $e = d^g$ fixes $E \setminus \{u_g\}$ and sends u_g to u_{2g}. Each be removes one further point of B while retaining u_{2g}. Hence $b^{g+1}e(be)^{s-1}$ resets Q. □

Taking $g = s - r - 1$ and $M = n - r - 1$ proves part (3) of Theorem 1.

4 The exact codimension-three maximum

Theorem 6 (Computer-assisted upper bound)**.** *For every $n \ge 5$ and $1 \le r < n - 3$,*

$$\mathcal{F}_{\mathrm{bin}}(n, n-3, r) = \mathcal{F}(n, n-3, r) = \min\{n - r, 9\}.$$

In fact, for $n \ge 6$ every $(n-3)$-subset in every strongly connected synchronizing automaton admits a compressing word of length at most nine, without any alphabet-size or image-reachability assumption. Consequently $\mathcal{F} = \mathcal{F}_{\mathrm{bin}} = 9 < U = 10$ whenever $n \ge 11$ and $1 \le r \le n - 10$.

Proof. An explicit lower bound. First assume $n \ge 10$. Put $L = n - 5 \ge 5$ and take $T = \{x_0, \dots, x_{L-2}, z\}$, $K = \{v, y, c, t\}$ and a further state d. Let b cycle T in the displayed order, cycle v, y, c, t, and fix d. Let a interchange z, y and c, d, send t to z, and fix the other states. Put $S = T \cup \{v, t\}$. For $r < L$, take the last r points of $x_0, \dots, x_{L-2}$ as A; for $r = L$, take $A = T$. The interchanges link the b-cycles and d, proving strong connectivity. Direct application gives $Q(abbbabaabb) = S$.

The following table lists the complements of all images reached within eight letters. The level is their minimum distance from S; each displayed successor has level at most one larger. For levels below eight both successors have $n - 3$ states. Induction proves that no word of length at most eight compresses S.

row	holes	level	a	b	row	holes	level	a	b
0	y, c, d	0	1	1	9	y, t, c	6	12	3
1	t, c, d	1	1	2	10	x_0, v, t	7	10	13
2	v, t, d	2	3	4	11	x_1, y, c	7	14	15
3	v, t, c	3	2	5	12	z, t, d	7	9	16
4	v, y, d	3	3	0	13	x_1, v, y	8	–	–
5	v, y, t	4	6	7	14	x_1, t, d	8	–	–
6	z, v, t	5	5	8	15	x_2, t, c	8	–	–
7	v, y, c	5	2	9	16	x_0, v, d	8	–	–
8	x_0, v, y	6	10	11					

The first possible missing T-states are z, x_0, x_1, x_2 at depths $5, 6, 7, 8$, attained respectively by $ababa\,b^j$ for $j = 0, 1, 2, 3$. Every other T-state remains present through depth eight. The word $ababbbab$ reaches row 16, whose image contains y, t; the next a merges them. Thus the indicated choices give $\mathrm{ac}(S, A) = \min\{n - r, 9\}$. In particular, at $n = 11$ either $A = \{x_3\}$ or $A = \{x_4\}$ gives length nine.

For synchronization, put $N = \mathrm{lcm}(L, 4)$. Then $e = a^2$ is the unit $t \mapsto y$, and b^N is the identity. Conjugates $b^{N-i}eb^i$ merge all opposite pairs in K. A d–K pair can be rotated to $\{d, v\}$ and then mapped by a to the opposite

pair $\{c, v\}$. The pair $\{c, z\}$ maps by a to $\{d, y\}$. The word b^4ab^{N-4} maps $\{d, z\}$ to $\{c, z\}$: $zb^4 = x_3$ exists since $L \geq 5$, and the final power fixes c and returns x_3 to z. Rotating a T-point to z therefore merges every d–T pair. Adjacent pairs in K rotate to $\{c, t\}$ and map by a to $\{d, z\}$. For a T–K pair, rotate its K-point to c: either the other point is z, or a gives a d–T pair. A T–T pair rotates one point to z and then a gives a T–K pair. Hence every pair merges. Repeatedly merging two points of the current image synchronizes Q.

Uniform upper bound. The following proof uses complete finite action certificates; the supplement contains all symbolic domain reductions and both independent enumerations. Put $P_h = \{E : |E| = 2,\ |Ew| = 1 \text{ for some } |w| \leq h\}$. A Frankl–Pin sequence consists of s-sets S_i and pairs $E_i \subseteq S_i$ with $E_i \nsubseteq S_j$ for $j < i$. The refinement of Kisielewicz, Kowalski and Szykuła [4, Theorem 1] gives compression length at most $10 - p(P_h) + h$ for $s = n - 3$. Any $q \leq 4$ distinct pairs admit a sequence of length q: choose each three-hole set outside its own pair, hitting an endpoint of every later pair, and fill the remaining holes. Therefore compression length ten forces $|P_h| \leq h$ for $h = 1, 2, 3$. Pair closure grows strictly until every pair is present in a synchronizing automaton; hence $|P_h| = h$ in this range. Write $P_h = \{E_1, \ldots, E_h\}$, with $E_1 = 01$. Each letter is a permutation or a rank-$(n-1)$ map with common kernel 01. Its inverse pairs of E_1 lie in P_2, and those of E_2 lie in P_3. There are precisely the nine ordered types in the table below.

Let $\mathcal{H}$ be the three-hole covers of P_3. Every image reached by depth six has holes in $\mathcal{H}$, or a further three letters compress before ten. The families R_i of images reached by depth i grow strictly before compression: a plateau would be invariant under all letters and could not be reset. Thus $|R_i| \geq i + 1$. A common $\mathcal{H}$-preserving domain D for tests of length at most L contains R_{6-L}, so $|D| \geq 7 - L$. The symbolic classifications in the supplement either contradict this count or reduce to a domain with $k = 7 - L$ images. All its images are then reached by depth $6 - L$, and every letter maps them into $\mathcal{H}$.

Project letters onto D and EXIT$=\mathcal{H} \setminus D$. Synchronization forces every vertex to exit. An initial exit distance k forces the k vertex distances to be a permutation of $1, \ldots, k$. For each such vector d, include *all* profiles p satisfying $d(i) \leq 1 + d(p(i))$, $d(\text{EXIT}) = 0$, and test whether all vertices can exit. This enlarged graph contains every actual alphabet with these distances. The complete checks have the following sizes; entries with several numbers correspond to separate domains.

type	E_2	E_3	domain sizes	profiles	vectors
0	02	12	4,4,6	11,38,61	24,24,720
1	02	03	no incoming edge to 0 from another state		
2	02	13	5	22	120
3	02	23	5,4	24,19	120,24
4	02	34	6,6	33,37	720,720
5	23	02	incompatible inverse pairs		
6	23	04	5	19	120
7	23	24	6,6,6;5,5,5,5	21,39,37;27,23,23,33	720 each;120 each
8	23	45	8	177	30240

All distance vectors fail, except three in type 0, one each in types 2,3,6 and two in one type 7 domain. For *every underlying row* of every allowed profile in each surviving vector, a five-state set is invariant. Strong connectivity excludes this for $n \geq 6$. Thus first exit has length at most $k - 1$, giving compression within $(k - 1) + L + 3 = 9$. One additional type 2 domain has four images and two immediate exits, so its exit distance is at most three, again giving $3 + 3 + 3 = 9$. Type 8 instead directly checks all 30240 possible finite distance vectors with initial distance at least seven; none survives, giving $6 + 3 = 9$.

The projections are uniform in n: only at most six named inputs matter, with the remaining distinct outputs renamed to a bounded supply of fresh labels. Every row preserving the required holes is enumerated; extra rows not extendible at a small n only enlarge the graphs. One implementation uses inverse-pair templates and reverse exit closure; the other enumerates literal forward partial rows and searches forward for exits. All profiles, row-closure flags and surviving vectors coincide. The supplement supplies the full classifications, exact domains, fresh-label bounds, source and data for all nine cases. This excludes compression ten and proves the uniform upper bound. Combining it with Lemma 2 and the lower construction gives the asserted local maximum for $n \geq 10$, using Theorem 1 when $r = n - 4$. For $5 \leq n \leq 9$, independently checked binary witnesses in the supplement attain $U = \min\{n - r, 9\}$ for every admissible r. □

5 A uniform two-block construction

Theorem 7. *Put $m = n - s \geq 3$ and $q = n - r \geq 2m + 2$. There is a binary strongly connected synchronizing automaton with reachable S attaining $\mathrm{ac}(S, A) = \min\{q, 3m - 1\}$. Consequently $\mathcal{F}_{\mathrm{bin}} = \mathcal{F} = U = q$ whenever*

$2m+2 \le q \le 3m-1$. *For larger q the construction gives only a lower bound.*

Proof. Write $t = m-1 \ge 2$ and $p = n-2t-1 \ge r+3$. Take $P = \{p_0, \dots, p_{p-1}\}$, $X = \{x_1, \dots, x_t\}$, $Y = \{y_1, \dots, y_t\}$ and one further state z. Letter a interchanges the last two P-states, fixes the others, and cycles $x_1, \dots, x_t, y_1, \dots, y_t, z$. Letter b sends p_0 to y_1, p_i to p_{i-1} for $i \ge 1$, increments indices in X, Y below t, sends both x_t, y_t to p_{p-1}, and fixes z. Put $S = P \cup X$, $A = \{p_0, \dots, p_{r-1}\}$ and $L = 3t+2$.

No compression before L. For $\rho_i = \min\{i, p-2\}$ define a symmetric potential Φ on unordered pairs, zero on singleton pairs, by the following table. Indices in X, Y run from 1 to t, and indices in P from 0 to $p-1$.

pair	potential
P–P, P–X, X–X (distinct)	L
$\{x_i, y_j\}$, $j = i$	$t-i+1$
$\{x_i, y_j\}$, $j > i$	$2t-i+1$
$\{x_i, y_j\}$, $j < i$	$L-j$
$\{x_i, z\}$	$2t-i+1$
$\{y_i, y_j\}$, $i < j$	$L-j$
$\{y_i, z\}$	$2t+1$
$\{p_i, y_j\}$	$\min\{L, \max\{L-j, 2t+3+\rho_i\}\}$
$\{p_i, z\}$	$\min\{L, 2t+2+\rho_i\}$

Substitution in the two letter actions gives

$$\Phi(E) \le 1 + \Phi(Ea), \qquad \Phi(E) \le 1 + \Phi(Eb).$$

Here are the boundary checks. The last-two-P interchange preserves ρ; a downward P-step decreases it by at most one. For P–Y pairs the a-boundary $y_t \to z$ decreases the second term from $2t+3+\rho$ to $2t+2+\rho$. The b-boundary $p_0 \to y_1$ with $j < t$ has source $L-j$ and target $L-j-1$, since $j \le t-1$; with $j = t$ its target is at least $L-1$. For X–Y, crossing x_t under a gives Y–Y with potential $L-j-1$ when $j < i$; crossing under b gives P–Y of at least that potential. The kernel pair $\{x_t, y_t\}$ has potential one. Crossing y_t in a Y–Y pair under a gives Y–z of potential $2t+1 = L-t-1$. For X–z, the a-boundary is $\{x_t, z\} \to \{y_1, x_1\}$, with values $t+1, t$; for Y–z, every a-image has potential $2t$. All remaining cases either retain potential, decrement an index by one, or enter a type whose potential is at least the source minus one. Clipping at L cannot increase a positive difference. Thus the inequalities hold for every index, not just tested sizes. Every distinct pair in S has potential L. Telescoping along a merging word proves its length is at least L; every compression merges an initial pair, giving the claimed lower bound.

No avoidance before q. Use singleton P-cells and t color cells, giving x_i, y_i color i and z color 1. Let T_a swap the last two P-cells and cycle the colors, and let T_b cycle $p_0, 1, \dots, t, p_{p-1}, \dots, p_1$. For the incidence matrix C both letters satisfy

$$D_\ell C = C T_\ell + f_z(e_1 - e_2).$$

These are the same row/column conventions and outer product as in (2). Since S has one state in every cell, expansion (3) applies, with coefficients counting trajectories at z. A trajectory from X needs at least $t+1$ steps to reach z; any entrance from P starts at y_1 after at least one step and then needs at least t more. Thus the first nonzero correction has position at least $t+2$. In the prescribed cell graph, color 2 needs $t-1$ letters to enter P at p_{p-1} and at least $p-r$ more to reach A. The only interchange is between adjacent last P-points and cannot shorten this descending path. A negative correction therefore reaches A only at length at least $(t+2) + (t-1+p-r) = p+2t+1-r = q$. Before then every protected coordinate has multiplicity at least one.

Attainment. After $a^{2t}b$, the missing P-point is p_{p-1}; after a further a it is p_{p-2}. This is the unique b^{p-r-1}-preimage of p_{r-1}. Hence $a^{2t}bab^{p-r-1}$ omits that protected point at length q. The $X \cup Y \cup \{z\}$ portion after $a^{2t}b$ is $\{x_2, \dots, x_t, y_1, z\}$. After a^t it contains x_t and y_t, so $a^{2t}ba^tb$ compresses at length L. The existence of x_t in the preceding set uses $t \ge 2$. Combining both upper witnesses with the lower bounds gives $\mathrm{ac}(S, A) = \min\{q, L\}$.

Structural requirements. Put $B = P \cup Y$, $h = p+t$ and $E = B \cup \{z\}$. Letter b cycles B in the order $p_0, y_1, \dots, y_t, p_{p-1}, \dots, p_1$, and $Qb^t = E$. The a-cycle connects all other states to B, proving strong connectivity. On E, $d = a^{2t}b^h$ fixes P, cycles $y_1 \to y_t \to \cdots \to y_1$, and sends z to y_t. Thus $e = d^t$ fixes B and sends z to y_1. Since $Ba^{t+1} = S$, the original-letter word b^tea^{t+1} reaches S. The word a^{2t+1} fixes $X \cup Y \cup \{z\}$ and transposes the last two P-points, adjacent on the b-cycle. These words generate every permutation of B, with inverses realized by positive powers. Moreover $Ba = (B \setminus \{y_1\}) \cup \{z\}$, $Bab = (B \setminus \{y_2\}) \cup \{z\}$, and $Babe = B \setminus \{y_2\}$. Permutations route any pair in a current image into the kernel of abe; repeated compression synchronizes B. Prepending b^te synchronizes Q. Finally $L = 3m-1 < \binom{m+2}{2}$ for $m \ge 3$, so meeting $q \le L$ determines both maxima as stated. □

6 The determined parameter region

Theorem 8. *Put $m = n - s$ and $q = n - r$. For all $1 \le r < s \le n$,*

$$\mathcal{F}_{\mathrm{bin}}(n,s,r) = \mathcal{F}(n,s,r) = \begin{cases} 1 & m = 0, \\ \min\{q, 3m\} & 1 \le m \le 3, \\ q & m \ge 4,\ q \le 3m-1. \end{cases}$$

The region $m \ge 4$, $q \ge 3m$ is not determined here.

Proof. The case $m = 0$ was proved above, and $m = 3$ is Theorem 6. For $m \ge 2$ and $q \le 2m + 1$, gaps one and two use the first two faces; larger gaps satisfy $n \ge 2s - r - 1$ and use Proposition 5. Together with the two-block band this gives q throughout $m \ge 3$, $q \le 3m - 1$. It remains to give the compression-range constructions for $m = 1, 2$.

For $m = 1$, $n \ge 4$ and $1 \le r \le n - 3$, let b cycle $u, v, z, x_0, \ldots, x_{n-4}$. Let a interchange z, u, send v to z and fix the other states. Put $S = Q \setminus \{u\}$ and choose A among the x_i. The possible holes through depth two are u, v, z: hole u goes to v under either letter, while hole v stays v under a and goes to z under b. Thus all words shorter than three are safe; bba compresses. Strong connectivity follows from b. The unit $e = a^2$ sends v to its b-predecessor u, so $e(be)^{n-2}$ resets Q by the cyclic-interval argument. Moreover $Qab^{n-1} = S$. The remaining $r = n - 2$ uses the first face.

For $m = 2$, $n \ge 7$ and $1 \le r \le n-6$, put $L = n-3 \ge 4$. Let b cycle $T = \{x_0, \ldots, x_{L-2}, z\}$ in that order and cycle w, v, u. Let a interchange z, u, send v to z and fix the other states. Put $S = T \cup \{v\}$ and choose $A \subseteq \{x_2, \ldots, x_{L-2}\}$. The complete hole sets through depth five are

row	0	1	2	3	4	5	6
holes	wu	wv	uv	zv	x_0u	x_0v	x_1w
level	0	1	2	3	4	5	5
a	1	1	3	2	5	$--$	$--$
b	1	2	0	4	6	$--$	$--$

Here wu means $\{w, u\}$, and similarly for the other entries. All these sets avoid A and all displayed transitions preserve $n - 2$ states. The word $ababb$ reaches row 6; the next a merges u, v. Thus $\mathrm{ac}(S, A) = 6$. The two b-cycles are linked by a, proving strong connectivity. Put $N = \mathrm{lcm}(L, 3)$, $e = a^2$, $f = b^{N-1}eb$ and $k = ef$. The units $e : v \mapsto u$ and $f : u \mapsto w$ make k collapse w, v, u to w, fixing T. Hence $Qkb = S$. The word $c = kafak$ sends w, v, u, z to w and fixes $T \setminus \{z\}$. Each bc removes the predecessor of z in the remaining cyclic interval, and $wbc = w$. Therefore $c(bc)^{L-1}$ resets Q. The cases $q \le 5$ were covered above, completing the proof. □

Reproducibility and scope. The 109-file supplement supplies the generators, independent lower checks, and complete case proofs for the upper bound, with standard-library Python and C++17 audits of sixteen domains and the matching cube. Finite controls check the symbolic reductions; they are not extrapolated. It verifies 290 binary reachable-image witnesses: all 286 triples with $2 \le n \le 12$, $(13, 8, 1)$, and $(13, 10, r)$ for $r = 1, 2, 3$. The proved upper bounds determine both maxima at all listed parameters. The complete finite frontier ends at $n = 12$; $(13, 9, 1)$ remains in $[11, 12]$. No alphabet separation or complete answer to Open Problem 4 is claimed.

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